



ELECT MAINT Department
SIMHADRI

Note No :1

Ref:SMTTP/ELECT MAINT/2024-25/EMLMIPTF/108850

Date:

08/05/2024(10:51:22)

Subject:REVISION OF LMI/OGN/OPS/ELEC/008 INSPECTION AND
PROTECTION OF WATER COOLED GENERATOR STATOR WINDING
PTFE HOSE CONNECTIONS.

LMIIs are reviewed periodically for further improvement and incorporation of Recommendation by internal or external audit committee and COS.

No changes were suggested by COS.

Put up for Periodical reviewing of the LMI.

May please approve.

Submitted by:

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On:08/05/2024(10:51:22) **Ref:**SMTTP/ELECT MAINT/2024-25/EMLMIPTF/108850

Note No:2

Put up for review plz (no changes)

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On:10/05/2024(17:40:01) **Ref:**SMTTPP/ELECT MAINT/2024-25/EMLMIPTF/108850

Note No:5

please modify LMI as standard LMI format as mentiond in LMI/OIN/GEN/002

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Note No:6

LMI modified as per new format.

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Note No:7

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Note No:8

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Note No:9

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Note No:11

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On:17/06/2024(23:10:17) **Ref:**SMTPP/ELECT MAINT/2024-25/EMLMIPTF/108850

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

**INSPECTION AND PROTECTION OF WATER
COOLED GENERATOR STATOR WINDING PTFE
HOSE CONNECTIONS**

LMI/OGN/OPS/ELEC/008 Rev:02

MAY 2024

Approved for Implementation by HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	INSPECTION AND PROTECTION OF WATER COOLED GENERATOR STATOR WINDING PTFE HOSE CONNECTIONS
2	LMI No.	LMI/OGN/OPS/ELEC/008 Rev:02
3	Reference Document	OGN-OPS-ELEC-008
4	Superseded Document	LMI/SMSTPS/OGN/OPS/ELEC/008 Rev:02 May 22
5	Prepared By	Name: Balabalaji Akula D G M (EMD)
6	Checked By	Name: Nageswara rao, DGM (EMD)
7	Reviewed By	Name: DK Tomar A G M (EM)-I/C
8	Reviewed By	Name: S.J.David G M (O&M)
	Approved By	Name: S.K.Sinha H O P (Simhadri)

TITLE	INSPECTION AND PROTECTION OF WATER COOLED GENERATOR STATOR WINDING PTFE HOSE CONNECTIONS
LMI No.	LMI/OGN/OPS/ELEC/008 Rev:02

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TITLE	INSPECTION AND PROTECTION OF WATER COOLED GENERATOR STATOR WINDING PTFE HOSE CONNECTIONS
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1.0 Introduction

Large capacity Turbo Generators using water cooling, number of failures of PTFE Hoses have been recorded. Objective of this LMI is to bring out the modalities with regards to the inspection of hoses, protection & hydraulic testing of PTFE Hoses during Generator Overhaul in order to avoid the in service failure of these hoses and consequent outage of machine.

2.0 Superseded Documents

NIL

3.0 Details of Revision

NIL

4.0 HYDRAULIC/LEAKAGE TEST DURING CAPITAL OVERHAULS

Inspection, care & testing technique are applicable while the machine has been dismantled and Rotor threaded out during major Generator Overhaul. Hydraulic Testing of the primary water system will be done by dry nitrogen at 6 kg/cm² pressure for 48 Hours *and any leakage shall be rectified before putting the Generator back in operation. However, sometimes it is necessary to do Helium gas detector test for detection of the minor leakages.*

During major overhauling of Generator when Rotor has been threaded out following would be done:-

D) INSPECTION OF PTFE HOSES: -

- i. **Damage to outside surface** – where major damage or cut like defects deeper than 1 mm are found, hose has to be changed.

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- ii. **Hose Alignment**-As far as is practicable, the physical location of the hoses should be checked for distortion. Check for distribution of bars (where hoses connection to winding are covered with insulation boxes, further stripping will not be done unless damage is evident). Major distortion or badly aligned hoses should be changed, in consultation with the Manufacturer.
- iii. **Circumferential spring back resistance** of the hose in non-connected condition should not exceed approximately 50 mm with respect to mounting position of screwed fitting. In case of higher spring back resilience, re-bending of the hoses is recommended. The hoses should be rebent and matched at a temperature of approximately 1500C which can be attained by industrial hot-blower.
- iv. **Crimping Connection** - Inspect crimp connection for double crimping, wherever not available it is to be changed accordingly.
- v. A visual inspection of Hoses should be carried out during major Overhauls to check whether any external contamination or deposits have been formed. Internal deposits could be due to fluctuations in water chemistry parameters for which corrective actions are to be taken. Teflon hoses with internal deposits are to be removed and a sample collected for further analysis.

E) PROTECTION OF PTFE HOSES DURING GENERATOR OUTAGE: Care has to be taken not to cause any mechanical damage to the hose during the opening of a Generator at the time of overhaul.

F) Protection of Open Pipes and Hoses: There have been instances where blockages have been found in pipework and hoses,

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which have led to loss of generation.

It is therefore recommended all site procedures are re-examined to ensure no plant is put into service without -

(e) Any replacement or refitted part having been checked for blockage before reinstallation. –

(f) Measures having been taken to ensure no debris enters the system during installation.

(g) A commissioning procedure having been followed which verified that all passage ways and pipelines were clear.

(h) Hot water flow test should be carried out during Capital overhaul and for each Teflon hose temperature is to be observed by concerned Engineer for ascertaining any partial/full chocking of the hollow conductors of stator links or Teflon hoses.

With regard to the stator cooling hoses, manufacturers normally fit protective caps but hoses with the insert type of protection can be supplied as spares and it is therefore further recommended that checks are made on spares held in stores and if any inserts are found they should be replaced by protective caps the hoses cannot be fitted with caps in place.

5.0 FREQUENCY OF CHECKING AND MEASUREMENTS:

Test/inspection shall be carried out during generator major overhauling.

6.0 The RESPONSIBILITY CENTERS FOR IMPLEMENTATION

Basic responsibility lies with respective Area in-charge

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7.0 EXCEPTION REPORTING & DISPENSATION OF DEVIATION

Exception will be reported to HOD (EM) and can be dispensed with Approval.

8.0 Review

HOD (EMD) will review & take the approval of competent authority (HOP). The document will be reviewed on 2 yearly basis or earlier, if required.

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9.0 Checklist:**CHECKLIST DURING OVERHAUL****UNIT****DATE**

S No	Description	Observation/Limits	Remarks
1	Any damage to outside surface of the hose	YES/NO If damage/cut deeper than 1mm is found, hose to be changed	
2	Condition of crimp connection		
3	Visual inspection of hose for any external contamination or deposits		
4	Hydraulic testing of primary water system and generator stator winding at 6 bar, 48 hours with N2 cushion	No leakage ,pressure drop is allowed	
5	Number of hoses changed		